



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to PAARVATHI FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

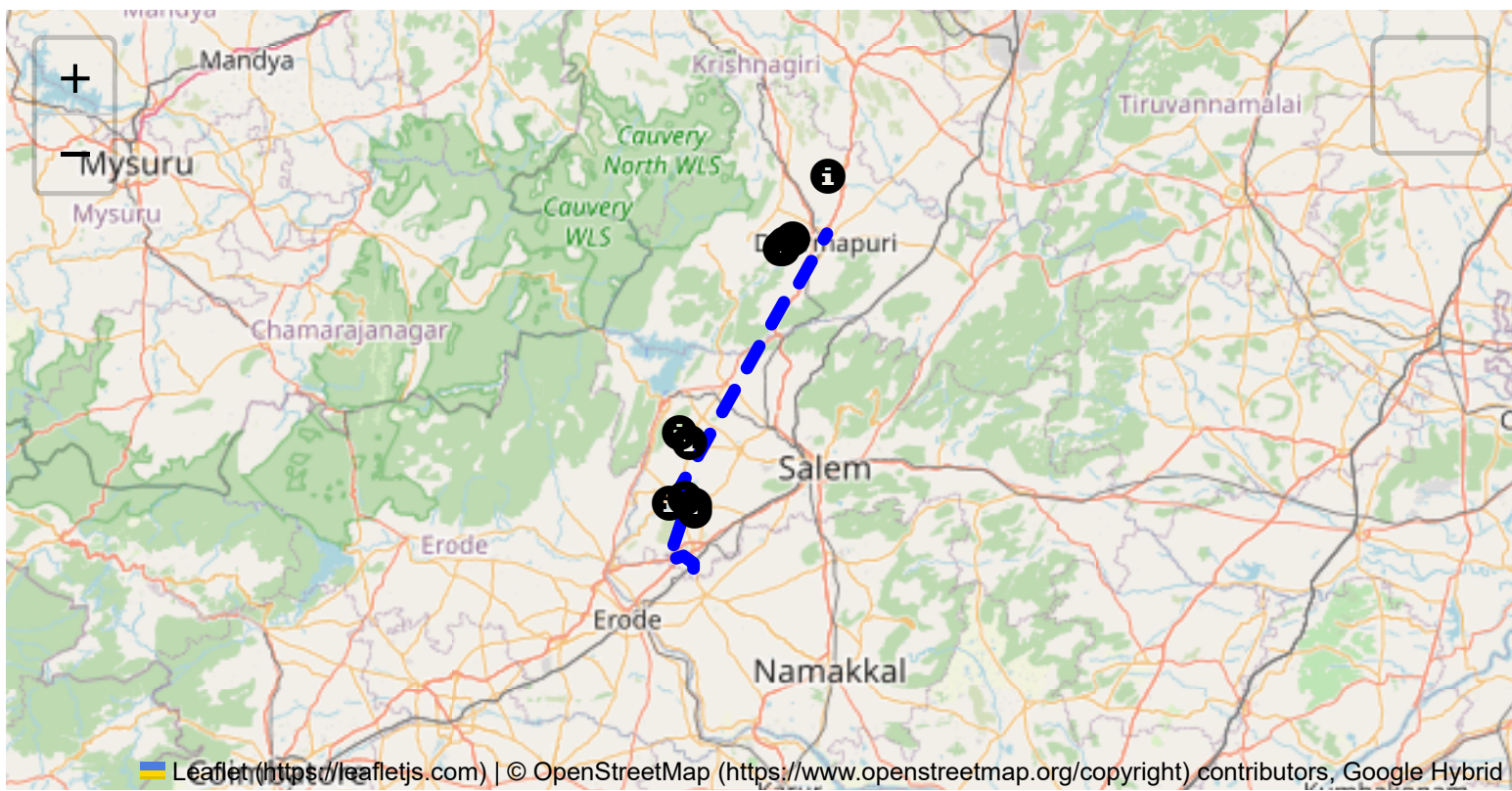
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

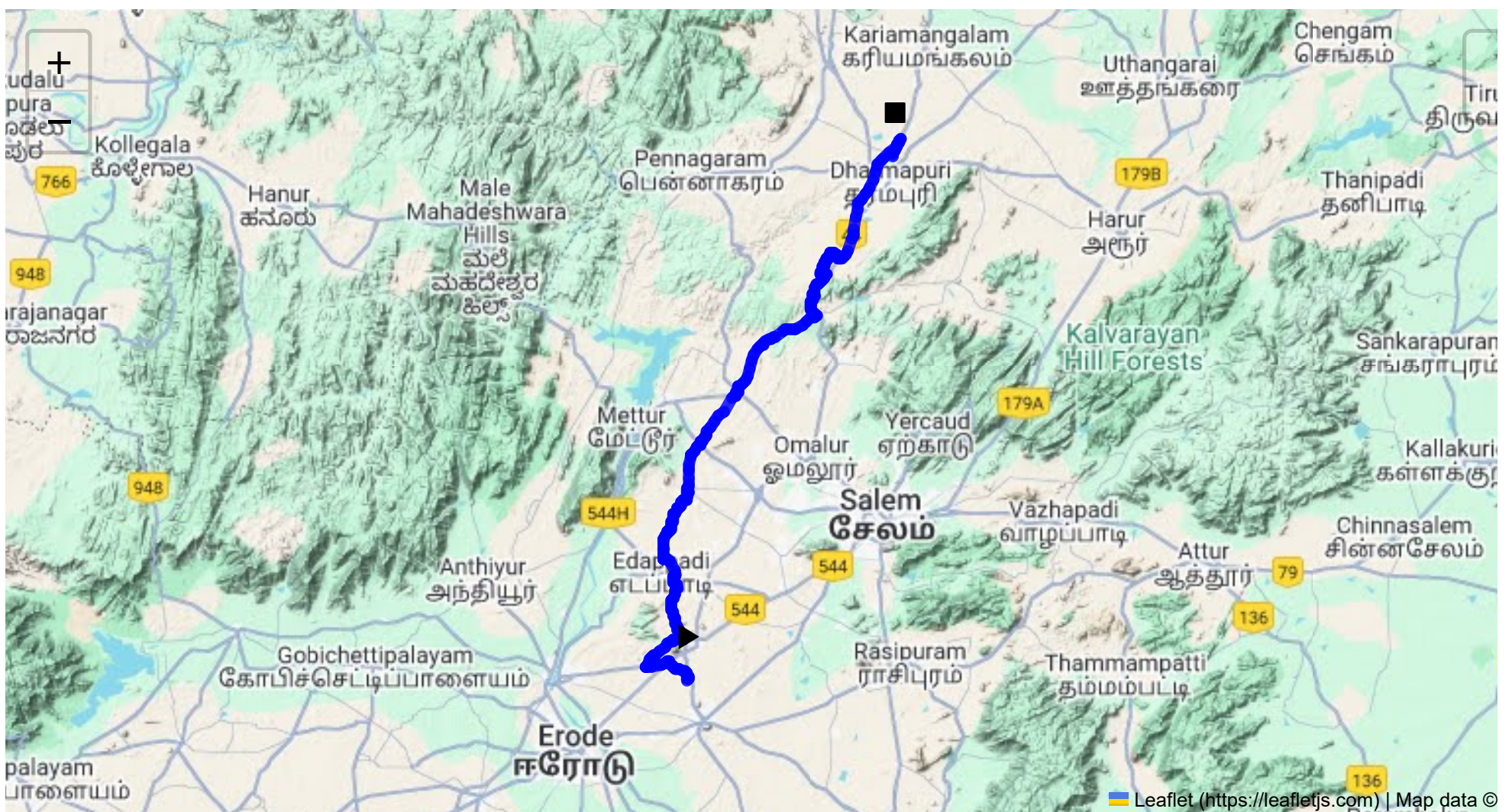
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 117.37 km
Estimated Duration: 2.7 hours
Adjusted Duration (Heavy Vehicle): 3.3 hours
Start: (11.4381, 77.8734)
End: (12.154, 78.163)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The specified route is approximately 117.37 kilometers in length, connecting Sangagiri to A.Reddihalli in Tamil Nadu. The route passes through multiple towns and villages, including Sankari RS, Sankari West, Padaiveedu, Kurumbapatti, Avanurkottai, Vellakkal, Maniathahalli, Dokkubodanahalli, Nallampalli, and MGR Nagar. The path mostly follows main roads that are key for regional connectivity.

2. Typical Weather Conditions and Potential Weather-Related Hazards

The region of Tamil Nadu experiences a tropical climate. Key weather considerations include:

- **Summer (March to June):** High temperatures up to 40°C, which could impact vehicle performance.
- **Monsoon (July to September):** Heavy rain leading to potential flooding, especially on rural roads with insufficient drainage.
- **Winter (November to February):** Mild weather, with occasional fog in the mornings, which can affect visibility.

3. Traffic Patterns

- **Peak Hours:** Morning (8:00 AM - 10:00 AM) and evening (5:00 PM - 8:00 PM) tend to be busiest, especially near town centers like Sangagiri and Sankari.
- **Congestion-Prone Areas:** Sankari RS - Erode Privu Rd and Bhavani Main Road in Sankari West frequently experience slowdowns. Roadworks can exacerbate congestion.

4. Assessment of Road Quality and Infrastructure

- **Road Conditions:** Generally well-maintained, but some stretches, particularly in more rural areas, may have potholes or uneven surfaces.
- **Infrastructure:** Presence of multiple junctions and level crossings which require cautious driving. Most areas have basic signage, but additional caution is required at night.

5. Suggestions for Alternative Routes for Emergencies

Consider using the Sankari to Bhavani road or NH44 if there are emergencies leading to major obstructions or hazardous conditions along the main route.

6. Local Regulations Affecting Hazardous Material Transport

Tamil Nadu imposes strict guidelines for the transportation of hazardous materials, including:

- Vehicle-specific permits & licenses
- Mandatory safety equipment (e.g., fire extinguishers, spill kits)
- Restricted times for transit through urban areas

7. Overview of Historical Incidents

There have been incidents along this route involving heavy vehicles due to:

- Overturned vehicles from sharp turns
- Accidents at busy junctions, largely due to misjudgment or speeding
- Inclement weather contributing to loss of control

8. Environmental Considerations and Sensitive Areas

The route crosses agricultural areas and occasionally near water bodies. Care should be taken to prevent contamination from accidental spills.

9. Analysis of Communication Coverage

- Communication Zones:** Well-covered near major towns and cities.
- Potential Dead Zones:** Parts of the route near Avanurkottai and Maniathahalli have weaker signals, which could potentially impede emergency communication.

10. Estimated Emergency Response Times

- Urban Areas (e.g., Sangagiri, Sankari):** Approximately 20-30 minutes due to better access to emergency services.
- Rural Areas (e.g., Vellakkal, Maniathahalli):** Can be up to an hour or more, dependent on road conditions and time of day.

12. Overall Summary of Risk Assessment

Overall, the route from Sangagiri to A.Reddihalli is moderately safe for transporting hazardous materials, provided that regulations are followed, and due caution is observed, especially during adverse weather conditions and peak traffic times. Better communication infrastructure in rural segments and enhanced road conditions in certain sections would further minimize risks.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58300, 77.85813	15 KM/Hr
1	Roundabout	High	11.60359, 77.83965	15 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	High	11.44900, 77.87406	15 KM/Hr
5	Turn	High	11.45352, 77.85698	15 KM/Hr
6	Turn	Medium	11.45764, 77.85219	30 KM/Hr
7	Turn	High	11.46303, 77.84945	15 KM/Hr
8	Turn	High	11.45571, 77.81702	15 KM/Hr
9	Turn	High	11.45615, 77.81677	15 KM/Hr
10	Turn	Medium	11.46916, 77.82431	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
11	Turn	Medium	11.49385, 77.86709	30 KM/Hr
12	Turn	High	11.49404, 77.86724	15 KM/Hr
13	Turn	Medium	11.49513, 77.86656	30 KM/Hr
14	Turn	Medium	11.50866, 77.86053	30 KM/Hr
15	Turn	Medium	11.52587, 77.85524	30 KM/Hr
16	Turn	Medium	11.52791, 77.85581	30 KM/Hr
17	Turn	Medium	11.55685, 77.85813	30 KM/Hr
18	Turn	Medium	11.56411, 77.85969	30 KM/Hr
19	Turn	High	11.57903, 77.85441	15 KM/Hr
20	Turn	Medium	11.57910, 77.85555	30 KM/Hr
21	Turn	High	11.58022, 77.85744	15 KM/Hr
22	Turn	Medium	11.58709, 77.85824	30 KM/Hr
23	Blind Spot	Blind Spot	11.60437, 77.83983	10 KM/Hr
24	Turn	Medium	11.61007, 77.84163	30 KM/Hr
25	Turn	Medium	11.61835, 77.84026	30 KM/Hr
26	Turn	Medium	11.64049, 77.84596	30 KM/Hr
27	Turn	Medium	11.64166, 77.84871	30 KM/Hr
28	Turn	High	11.67319, 77.86041	15 KM/Hr
29	Turn	Medium	11.67365, 77.86215	30 KM/Hr
30	Turn	High	11.68487, 77.86933	15 KM/Hr
31	Turn	High	11.68478, 77.87285	15 KM/Hr
32	Turn	High	11.69463, 77.87589	15 KM/Hr
33	Turn	High	11.69549, 77.87444	15 KM/Hr
34	Turn	High	11.76020, 77.89071	15 KM/Hr
35	Turn	High	11.75960, 77.89221	15 KM/Hr
36	Turn	Medium	11.83330, 77.94432	30 KM/Hr
37	Turn	Medium	11.83370, 77.94485	30 KM/Hr
38	Turn	Medium	11.84179, 77.94732	30 KM/Hr
39	Turn	Medium	11.89747, 77.98079	30 KM/Hr
40	Turn	Medium	11.90011, 77.98213	30 KM/Hr
41	Turn	Medium	11.91901, 78.02623	30 KM/Hr
42	Turn	High	11.93618, 78.05392	15 KM/Hr
43	Turn	Medium	11.93710, 78.05390	30 KM/Hr
44	Turn	High	11.95824, 78.04981	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
45	Turn	Medium	11.95992, 78.04947	30 KM/Hr
46	Turn	Medium	11.96320, 78.05139	30 KM/Hr
47	Turn	High	11.96401, 78.05134	15 KM/Hr
48	Turn	Medium	11.96504, 78.04975	30 KM/Hr
49	Turn	High	11.96597, 78.04920	15 KM/Hr
50	Turn	Medium	11.96679, 78.04942	30 KM/Hr
51	Turn	Medium	11.96804, 78.05080	30 KM/Hr
52	Turn	High	11.99256, 78.06763	15 KM/Hr
53	Turn	Medium	11.99302, 78.06753	30 KM/Hr
54	Turn	Medium	11.99369, 78.06658	30 KM/Hr
55	Turn	Medium	11.99394, 78.06400	30 KM/Hr
56	Turn	Medium	11.99311, 78.06224	30 KM/Hr
57	Turn	High	11.99371, 78.06005	15 KM/Hr
58	Turn	Medium	11.99530, 78.06006	30 KM/Hr
59	Blind Spot	Blind Spot	12.00348, 78.06524	10 KM/Hr
60	Turn	High	12.00421, 78.06375	15 KM/Hr
61	Turn	High	12.00788, 78.06535	15 KM/Hr
62	Turn	High	12.00782, 78.06596	15 KM/Hr
63	Turn	Medium	12.00916, 78.06688	30 KM/Hr
64	Turn	High	12.01969, 78.07389	15 KM/Hr
65	Turn	High	12.02125, 78.07316	15 KM/Hr
66	Blind Spot	Blind Spot	12.02364, 78.07523	10 KM/Hr
67	Turn	Medium	12.02321, 78.07559	30 KM/Hr
68	Turn	Medium	12.02271, 78.07832	30 KM/Hr
69	Turn	Medium	12.02144, 78.08161	30 KM/Hr
70	Turn	Medium	12.02091, 78.08188	30 KM/Hr
71	Turn	High	12.02034, 78.08261	15 KM/Hr
72	Turn	Medium	12.01879, 78.08238	30 KM/Hr
73	Turn	High	12.01399, 78.08651	15 KM/Hr
74	Blind Spot	Blind Spot	12.17892, 78.17243	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Parusuraman Hospital	11.697164, 77.877792	30 km/h	Medium
2	hospital	JP Nature Cure Hospital Siddha and Naturopathy	11.7002949, 77.8770897	30 km/h	Medium
3	hospital	SSS Hospital	11.7501471, 77.8832809	30 km/h	Medium
4	clinic	DSR Clinic	11.7603176, 77.8908853	30 km/h	Medium
5	hospital	Srilakshmi Hospital	11.7601, 77.892272	30 km/h	Medium
6	hospital	Government Hospital	11.758158, 77.894528	30 km/h	Medium
8	hospital	Child Hospital	11.833001, 77.943415	30 km/h	Medium
9	hospital	Vasu Speciality Hospital	11.8331097, 77.943169	30 km/h	Medium
10	hospital	Rajamanickam Hospital	11.8332408, 77.9431081	30 km/h	Medium
11	hospital	Naveen Nursing Home	11.8335355, 77.9437441	30 km/h	Medium
12	hospital	Sri Sakthi Hospital	11.835714, 77.9451014	30 km/h	Medium
13	clinic	Sri Krishna Clinic	11.835443, 77.9450587	30 km/h	Medium
14	hospital	Selvakumar Hospital	11.836215, 77.945842	30 km/h	Medium
15	clinic	SK Clinic	11.83619, 77.9457752	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	college	Kavery Educational Institutions	11.8255319, 77.9342256	30 km/h	Medium
16	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium

Route Photos of Risky Spots

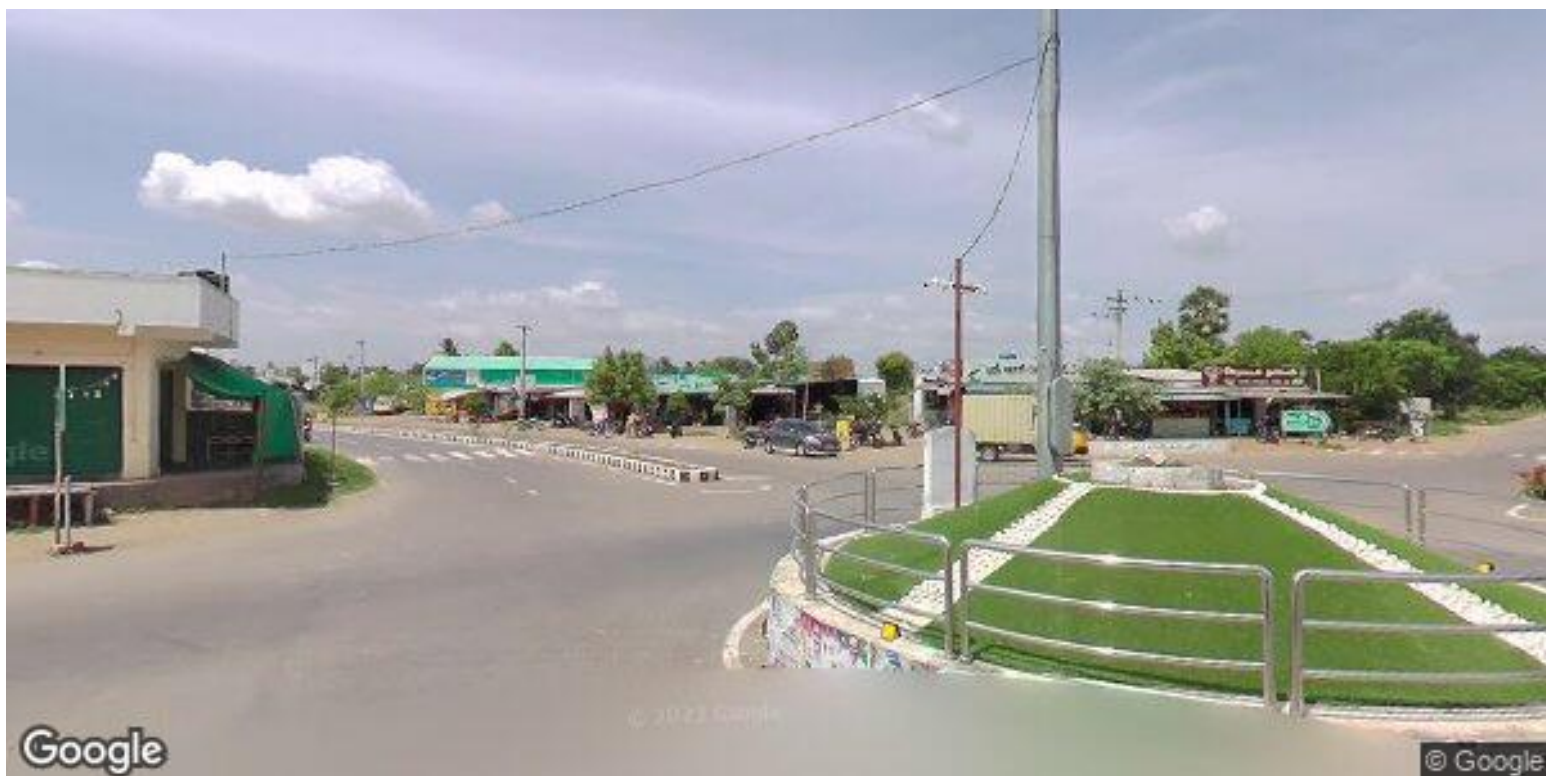


Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58300, 77.85813



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.60359, 77.83965



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45764, 77.85219



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45571, 77.81702



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45615, 77.81677



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46916, 77.82431



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49385, 77.86709



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49404, 77.86724



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49513, 77.86656



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.50866, 77.86053



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52587, 77.85524



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52791, 77.85581



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55685, 77.85813



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56411, 77.85969



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.57903, 77.85441



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57910, 77.85555



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58022, 77.85744



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58709, 77.85824



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.60437, 77.83983



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61007, 77.84163



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.61835, 77.84026



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64049, 77.84596



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64166, 77.84871



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67319, 77.86041



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67365, 77.86215



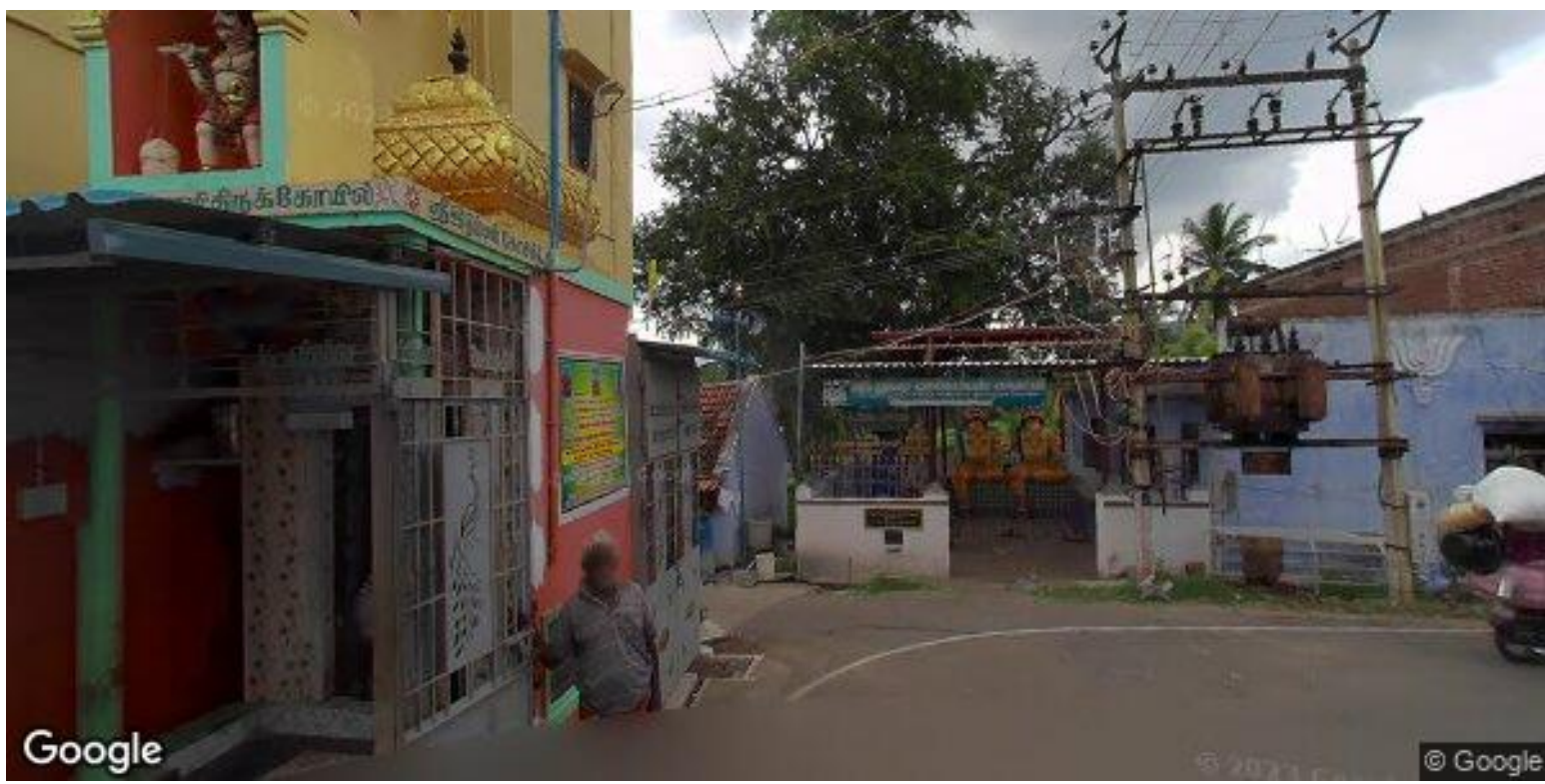
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.68487, 77.86933



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.68478, 77.87285



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69463, 77.87589



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.69549, 77.87444



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.76020, 77.89071



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.75960, 77.89221



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.83330, 77.94432



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.83370, 77.94485



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.84179, 77.94732



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.89747, 77.98079



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.90011, 77.98213



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91901, 78.02623



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.93618, 78.05392



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93710, 78.05390



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.95824, 78.04981



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95992, 78.04947



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96320, 78.05139



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96401, 78.05134



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96504, 78.04975



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96597, 78.04920



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96679, 78.04942



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96804, 78.05080



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.99256, 78.06763



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.99302, 78.06753



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.99369, 78.06658



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.99394, 78.06400



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.99311, 78.06224



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.99371, 78.06005



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.99530, 78.06006



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 12.00348, 78.06524



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.00421, 78.06375



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.00788, 78.06535



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.00782, 78.06596



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.00916, 78.06688



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.01969, 78.07389



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.02125, 78.07316



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 12.02364, 78.07523



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.02321, 78.07559



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.02271, 78.07832



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.02144, 78.08161



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.02091, 78.08188



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.02034, 78.08261



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.01879, 78.08238



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.01399, 78.08651



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 12.17892, 78.17243

